MIL-R-22930A (WEP)
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SUPERSEDING
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7 JUNE 1961

MILITARY SPECIFICATION

RADIO SET AN/ARC-88A(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide radio frequency reception and transmission facilities for data communications systems such as the Digital Data Communication Set AN/ASW-21. Operationally, the equipment shall provide two-way communication of digital data information on a time division basis by means of frequency shift keyed type of modulation.

*1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Receiver-Transmitter	RT-748A(*)/ARC-88A	3.4.1
Control, Radio Set	C-6362(*)/ARC-88A	3.4.2
Mounting	MT-2653(*)/ARC	3.4.3
Cooler, Air, Electronic Equipment	HD-513(*)/ARC	3.4.4
Indicator, Standing Wave Ratio	IM-178(*)/ARC-88	3.4.5

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For Use With Electronic Equipment in Aircraft)
MIL-C-26482	Connector, Electric, Circular, Miniature, Quick Disconnect

FSC 5821

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MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference, Control Requirements, Aircraft Equipment
MIL-L-6723	Lights, Aircraft, General Specification for
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-C-8384	Connectors, Plug and Receptacle, Electrical (Molded Body) and Accessories, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

PUBLICATIONS

ARINC No. 410, "Mark 2 Standard Frequency Selection System"

*2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCI, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

(2) Copies of ARINC Specification No. 410 may be obtained from ARINC, 1700 K St., N. W. Washington, D. C.

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3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6..3)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors, and diodes shall be chosen and applied, and the complements reported, as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

*3.1.3 Switches - Rotary switches mechanically positioned by other than manual means shall be mounted by standoffs.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

*3.2.2 Total Weight - The total weight of the equipment excluding cables, shall be a minimum consistent with good design and shall not exceed 34.6 pounds.

*3.2.3 Operating Life -

*3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 100 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

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* 3.2.3.3 Reliability In Mean Time Between Failures (MTBF) - The equipment shall have 150 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

* 3.2.4 Time Totalizing Meter - The receiver-transmitter shall contain a time totalizing meter in accordance with Specification MIL-M-7793.

3.2.5 Cabling and Connection -

3.2.5.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.5.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.6 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

* 3.2.7 Interference Control - The generation of radio interference by the radio set and the vulnerability of the radio set to radio interference shall be controlled within the limits of Specification MIL-I-6181 with the following exceptions:

(1) Antenna Conducted Keyup Condition - Spurious output between 200 mc and 500 mc shall not exceed 100 microvolts average and 400 microvolts maximum.

(2) Antenna Conducted Keydown Condition - The magnitude of all radiated signals from 200 mc to 500 mc, inclusive, on other then the carrier frequency shall be 40 db or more below the carrier signal. The magnitude of all radiated signals below 200 mc shall be 70 db or more below the carrier signal. The magnitude of all radiated signals above 500 mc shall be 80 db or more below the carrier signal.

(3) Receiver Intermodulation - The magnitude of the receiver intermodulation test signals shall be at least 80 db above one microvolt at the test sample terminal.

(4) Power Line Conducted Interference and Susceptibility - Frequency limits shall be 150 kc to 100 mc when testing the equipment. For power line conducted interference the limits at 25 mc shall apply over the 25mc to 100 mc range. Power line conducted interference shall not exceed 20 db above Specification MIL-I-6181 steady state limits when the transmitter is keyed at a 12 cps switching rate.

(5) Radiated Interference - An increase of 16 db above the level stipulated in Specification MIL-I-6181 will be allowed in the frequency range from 80 mc to 1350 mc. The carrier output frequency may exceed this 16 db limitation by an additional 30 db.

(6) Images and Other Spurious Responses - When receiving, the receiver shall attenuate images and other spurious responses to the maximum practicable extent and in any case not less than 60 db relative to the desired signal.

(7) IF-Rejection - Under either standard or service conditions, the receiver shall present an attenuation of not less than 80 db, relative to the response

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of the desired signal to signals on the frequencies of the IF amplifiers.

* 3.2.8 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.2.9 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance, except that only the antenna input and power connectors shall be mounted on the face of the equipment.

*3.2.10 Overload Protection - A 5 ampere non-temperature sensitive fuse shall be provided in the DC primary power circuit.

*3.2.11 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified in MIL-STD-704 for a period of at least 15 minutes, and shall automatically resume normal operation when the voltage returns within limits.

3.2.12 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.13 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (30° ±10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 VAC and/or 27.5 ± 0.5 VDC

3.2.14 Service Conditions - The equipment shall operate under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment except as modified herein.

*3.2.14.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves II and IV of Specification MIL-E-5400. Satisfactory performance when normally mounted (with vibration isolators in place) shall be demonstrated by compliance with the service condition requirements as specified herein. Satisfactory performance when mounted rigidly (without vibration isolators in place) shall be demonstrated by lack of mechanical failures and satisfactory operation under service condition limits after completion of the vibration test.

*3.2.14.2 Altitude - The equipment shall operate satisfactorily at an altitude of a least 70,000 feet when supplied with external cooling air per Table 1, and 50,000 feet when supplied with ambient air from an attached blower. For the latter case, the ambient air temperature shall be as specified in Specification MIL-E-5400. Cooling air pressure drop of the radio set shall be approximately as shown in Table 2.

TABLE 1

Cooling Air Requirements - AN/ARC-88A cooling air requirements at maximum system transmit-receive duty cycle in +71°C ambient air temperature when supplied from aircraft cooling system is:

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TABLE 1 (Continued)

Air Temperature °C	Flow Rate, lbs/hr
-15	27.2
- 5	27.5
5	29.0
15	32.3
25	37.8
35	46.4
45	58.0

TABLE 2

Cooling Air Pressure Drop - Cooling air pressure drop of Radio Set AN/ARC-88A in nominal values is:

Air Flow, lbs/hr	Pressure Drop, Inches of Water (Ref. Density= 0.0765 lb. per cu. ft.)
24	0.051
40	0.099
60	0.170
80	0.250
100	0.344

*3.2.14.3 Acoustic Noise Test - No malfunction of the equipment shall occur when subjected to a jet engine exhaust noise of 120 db relative to 0.0002 dyne per square centimeter at frequencies between 20 and 10,000 cps.

3.2.15 Pressurization - Items of equipment to be sealed shall comply with Specification MIL-E-5400 except that a safety valve is not required, mechanical pressurization pumps shall not be provided in the equipment and the cases shall be designed to be pumped with dry air to a differential of 2 to 4 pounds per square inch at sea level pressures and room ambient temperatures. Loss of pressure shall not exceed 5 pounds per square inch for a 24 hour period at 50,000 feet altitude and +55°C ambient temperature.

*3.2.15.1 Air Pressure Indicator - A non-air-release type pressure indicator shall be mounted in the front of the cover of the receiver-transmitter and shall give a visual indication when the internal air pressure is adequate for proper operation of the equipment.

*3.2.16 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts. Application of primary power to the equipment will be controlled by the associated digital data communications system.

(1) AC Power (Three Phase) 115/200V, Category "B", 200 VA

(2) DC Power 28V, Category "B", 4 Amps

3.2.17 Warm-Up-Time - The warm-up time for the equipment shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

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*3.2.18 Filament Protection - A protective device shall be incorporated in the equipment using zener diodes to maintain the filament voltage of the electron tubes at the proper operating levels when circuits may become unbalanced due to electron tube burnouts and module removals.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply under standard conditions. Reduced performance and service requirements shall apply only where specifically outlined. Equipment performance under service conditions shall consist of a duty cycle wherein the equipment is operated on a continuous cycle of 5 seconds reception and 1 second transmission. A duty cycle consisting of 10 minutes reception and 5 minutes transmission may be used during performance measuring periods.

3.3.2 Frequency Channels - By means of control box selection, it shall be possible to select any frequency in the range of 225.0 to 399.9 mc in 0.1 mc increments. These main channel frequencies shall correspond with the arithmetic series 225.0, 225.1, 225.3------399.9 mc.

3.3.3 Terminations -

3.3.3.1 Receiver RF - RF input to the receiver shall be through a 52 ohm cable from a signal generator having an effective source impedance of 52 ohms. Sensitivity shall be in terms of "open-circuit" microvolts.

3.3.3.2 Transmitter RF - The RF output load (Dummy Antenna) shall consist of a 52-ohm noninductive load which shall include means suitable for measuring the RF power output.

3.3.3.3 Receiver Output - The data output of the receiver shall be terminated in a 2000-ohm noninductive load.

3.3.3.4 Transmitter Data Input - The data input circuit of the transmitter shall present a load of not less than 2500 ohms to the signal source.

3.3.3.5 Turn-around and Carrier-on Pulse Input Terminations -

(a) Turn-around Input - The transmitter turn-around input circuit shall present a load of not less than 100,000 ohms to the keying source.

(b) Carrier-on Input - The carrier-on input circuit of the transmitter shall present a load of not less than 100,000 ohms to the keying source.

3.3.4 Receiver Sensitivity - Sensitivity of the receiver under standard conditions shall be such as to produce 20 db of quieting for a -98 dbm (5.6 open circuit microvolts) input signal. Quieting shall be measured at the data output terminals through a low pass filter providing attenuation of 3 db at 10 kc. A signal level of not more than -85 dbm shall produce 20 db of quieting under service conditions.

3.3.5 Selectivity - The overall selectivity of the receiver shall fall within the limits set forth below for either standard of service conditions:

Attenuation Below Resonant Response	Total Bandwidth
6 db	90 kc min.
60 db	200 kc max.

Peak to valley ratio shall be not more than 3 db over the 90 kc bandwidth.

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3.3.6 Regeneration - The interstage and overall regeneration of the receiver shall be so minimized that no evidence of instability, unnecessary oscillation, or excessive regeneration shall occur under either standard or service conditions.

3.3.7 Receiver Overall Frequency Responses - The response shall not vary more than 5% from the maximum data signal over a frequency range from dc to 7,000 cps with sine wave frequency modulation and ± 20 kc deviation. Test equipment frequency linearity error shall not be included in the 5% limit.

3.3.8 Receiver Output Linearity - The data output voltage shall be linear within ± 5% of the maximum peak-to-peak value data output voltage for signal deviation up to ± 20 kc.

*3.3.9 Receiver Amplitude Variation - With a constant data input signal at ± 20 kc deviation at a 1000 cps rate, the output shall not vary more than 10% for RF signal level variation from limiter saturation to 20 db above this level. (Limiter shall be saturated with signal input to the receiver of -80 dbm.)

3.3.10 Receiver Output Capability - The receiver shall be capable of delivering an output of at least 10 volts peak-to-peak into the data output load with a signal having an RF level sufficient for limiter saturation and frequency modulated sinusoidally at ± 20 kc deviation with a 1000 cps rate.

3.3.11 Receiver Internal Noise Level - The internal noise measured at the data output terminals through the low pass filter described in 3.3.4 shall not exceed 5% of the peak-to-peak data signal when an unmodulated RF signal of -67 dbm (200 open circuit microvolts) is applied to the receiver.

*3.3.12 RF Power Output - The transmitter shall be capable of delivering the following values of R F carrier energy within the tuning range:

Standard Conditions	20 watts minimum
Service Conditions	7 watts minimum
Reduced Performance Test at 102 vac and 22 vdc	3 watts minimum

3.3.13 Modulation Characteristics - The transmitter shall meet the following performance requirements for modulation when operated under both standard and service conditions.

3.3.13.1 Type of Modulation - The transmitter shall employ frequency shift modulation.

3.3.13.2 Capability - The modulation capability shall be such that +5.0 volts applied to the data input terminals shall increase the carrier frequency 20 kc (± 10%) and -5.0 volts applied to the data input terminals shall decrease the carrier frequency 20 kc (± 10%).

3.3.13.3 Transmitter Noise Level - Without modulation, the frequency deviation developed by internal noise shall not exceed 5% of the maximum peak-to-peak deviation of the data signal.

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3.3.13.4 Transmitter Frequency Linearity - The FSK modulated RF frequency shall be linear within ±5% of the maximum peak-to-peak frequency deviation over a range of data input voltage levels between 0 volts and ±5.0 volts.

3.3.13.5 Transmitter Frequency Response - The peak-to-peak frequency deviation shall not vary more than 5% with a sine wave modulated signal applied to the data input terminals of constant level and over the frequency range from 0 to 7000 cps. Test equipment frequency linearity error shall not be included in the 5% limit.

3.3.14 Frequency Stability - The frequency stability of the complete equipment after operation under either standard or service conditions for 5 minutes or more shall be such as to maintain the equipment frequency within the limits shown below provided the external ambient temperature does not exceed 55°C.

Transmitter Carrier Frequency Tolerances	±7 kc
Receiver Resonant Frequency Tolerance	±7 kc

3.3.15 Channel Changing Cycle - The time elapsing between the instant the channel selector switch is positioned to select a given channel and the instant the equipment is in an operative condition on the selected channel shall not be greater than 6 seconds under service conditions. This requirement shall hold for shifting the operating frequency of the equipment between any two available channels.

*3.3.16 Accuracy of Automatic Tuning - When the same frequency channel is selected several times in succession, the transmitter RF power output and the receiver output power (with an input signal of -75 dbm or greater with a 20 kc deviation at 100 cps) shall not vary by more than 0.5 db from the power outputs obtained initially. This requirement shall be met under standard or service conditions provided such conditions are static.

3.3.17 Cable Length - Under the standard or service conditions, the equipment shall operate as required by this specification when using control cables up to 25 feet in length fabricated from #22 AWG cooper wire, or when using cables up to 110 feet in length fabricated from #16 AWG wire.

3.3.18 Receive to Transmit Switching Requirements - The equipment shall meet the following requirements for switching from receive to transmit operation.

3.3.18.1 Receive to Turn-around Operation - The equipment shall remain in receive condition when -13 to -30 volts is applied to the "Turn-around" input circuit. Application of a "Turn-around" pulse of 0 to -2 volt level shall cause all applicable transfer relays within the equipment to be operated to the transmit position.

3.3.18.2 Turn-around Keying Time - Turn-around operation, from receive to prepared to transmit condition, shall not exceed 28 milliseconds after applications of the "Turn-around" pulse to the input circuit. Turn-around operation from prepared to transmit to receive shall not exceed 28 milliseconds.

3.3.18.3 Leakage Power - Leakage RF power from the transmitter shall not exceed -85 dbm at the operating frequency when in "Turn-around" condition.

3.3.18.4 Carrier-on Operation - From "Turn-around" condition, the transmitter shall be capable of being keyed upon application of a keying voltage having the following characteristics. The magnitude of the keying pulses shall be between 13 and 25 volts peak-to-peak, maximum deviation of the voltage shall be within +10 and -30 volts. The carrier will be on during the most positive portion of the voltage.

3.3.18.5 Carrier on Keying Time - 80% of the specified RF power level shall be applied to the output load within 160 microseconds after application of each "Carrier-on" pulse. The RF power in the output load shall decrease to not more than -85 dbm within 80 microseconds after cessation of the "Carrier-on" pulse.

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*3.3.19 Frequency Selection System - The equipment shall use the 2-out-of-5 code system in accordance with ARINC Specification No. 410 "Mark 2 Standard Frequency Selection System" dated 1 October 1961.

3.4 Detail Requirements -

*3.4.1 Receiver-Transmitter RT-748/ARC-88A - The receiver-transmitter shall meet the following requirements.

3.4.1.1 Function - The receiver portion of the receiver-transmitter shall receive frequency shift keyed RF energy from the antenna, amplify and convert it to digital type information to be applied to data multiplex equipment. The transmitter portion shall generate RF power, frequency shift keyed by digital information supplied in time division sequence from data multiplex equipment, and transmit it to the load specified herein.

*3.4.1.2 Form Factor and Mounting - The receiver-transmitter shall be contained in a single sealed case which meets the applicable requirements of Specification MIL-C-172 and capable of being mounted in an aircraft by means of Mounting MT-2653/ARC. It shall be possible to dismount the unit and remove it from its case using ordinary hand tools and without requiring removal of any cables other than the coaxial and power connectors. Vibration isolators shall be arranged to furnish all support required by the receiver-transmitter and shall provide space between the mounting rails for the external air ducts which are a part of the sealed case assembly. The maximum dimensions of the receiver-transmitter shall not exceed 7 inches high by 9-9/16 inches wide by 15-1/8 inches long.

*3.4.1.2.1 Warning Notice and Mechanical Clearance - The following notice shall be placed on the plenum assembly: "Mounting surface of modules must be flush with chassis before tightening mounting screws. Adequate clearance shall be provided to allow for ease of adjustment of all circuits during electrical alignment procedures."

*3.4.1.3 Contents - The receiver-transmitter shall contain the complete transmitter, FSK modulator, receiver, direct crystal-controlled frequency generator, frequency selection system, automatic tuning devices, and all other parts to make the equipment function except the radio set control. It shall be possible to remove or replace any plug-in unit located on the main chassis using ordinary hand tools and without unsoldering any connections. It shall be further possible to replace any plug-in unit with another without resynchronization except that realignment of tuning elements may be necessary. An additional wire shall be connected between the external connector and the appropriate module to permit application of an external bias to desensitize the receiver for test purposes.

*3.4.1.4 Weight - The weight of the receiver-transmitter shall not exceed 28.2 pounds.

3.4.1.5 Electrical Connections -

*3.4.1.5.1 External - The following electrical connections shall be provided on the casting of the receiver-transmitter:

(1) Coaxial connector Type N, for connection of receiver-transmitter to the antenna circuits.

(2) Two connectors conforming to Specification MIL-C-26482 shall be provided for connection of power and control circuits.

(3) A coaxial connector, Type N, shall be used on the output of the reflectometer-filter.

*3.4.1.5.2 Internal - Module power and signal connectors shall be of a quality conforming to the specification requirements of Specification MIL-C-8384 except that contact retention shall be 1 oz. minimum and 8 oz. maximum. Coaxial connectors shall meet the performance requirements of this specification and may be an integral part of the module power and signal connectors.

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3.4.1.6 Internal Controls - The receiver data output control shall be accessible with the unit removed from its case.

3.4.1.7 External Controls - The following controls shall be located on the front of the receiver-transmitter.

(1) Receiver zero reference control.

(2) Transmitter data input level control.

(3) Momentary contact switch for "Turn-around" test.

(4) Momentary contact switch for "Carrier-on" test.

*3.4.2 Control, Radio Set C-6362/ARC-88A - This control shall conform to the applicable requirements of Specification MIL-C-6781 and shall be suitable for installation in a console panel.

*3.4.2.1 Function - This unit shall provide prime control of the AN/ARC-88A Radio Set and shall include all controls for operation and frequency selection of Receiver-Transmitter RT-748/ARC-88A. It shall be possible to manually select any of the availabe frequencies and preset up to 20 channels on any of these available frequencies. The unit shall have red lighting within the plastic edge-lighted plate in accordance with Specification MIL-P-7788.

*3.4.2.2 Form Factor and Mounting - The over-all dimensions shall not exceed 5-3/4 inches in width, 4-7/8 inches in length and 6-1/4 inches in depth.

*3.4.2.3 Weight - The weight of the control, including the mating connector, shall not exceed 3.7 pounds.

*3.4.2.4 Contents - The control shall contain all circuits necessary to control manual frequency or preset channel selection.

*3.4.2.5 Electrical Connections - All electrical connections shall be made through a connector conforming to Specification MIL-C-26482.

*3.4.2.6 Controls, Panel-Mounted - The following controls shall be located and suitably marked on the front panel.

(1) A 20-position rotary switch-type control "CHAN" for selecting any one of 20 predetermined main frequencies.

(2) Three controls operating a digital display "MEGACYCLES" for manually selecting any one of 1750 channels. The frequency readout display shall indicate the frequency on which the equipment is operating when in the "MAN" position.

(3) A 2-position switch to allow rapid selection of two operating frequencies as follows: Position 1- "PRESET", for operation on any one of the predetermined frequencies selected by (1) above; position 2-"MAN", for operation on any of the manually selected frequencies determined by (2) above.

(4) Means for readily setting up new predetermined frequencies on the 20-position switch type control.

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*3.4.2.7 POWER-ON Indicator Light - A yellow light on the panel shall be turned on when the radio set is turned on. The words "POWER ON" shall be located below the light. The light shall conform with the requirements of Specification MIL-L-6723.

*3.4.3 Mounting MT-2653/ARC - The mounting shall meet the following requirements:

3.4.3.1 Function - The mounting shall provide a means of securing the receiver-transmitter to a plane surface within the aircraft or vehicle.

*3.4.3.2 Form Factor and Mounting - The mounting shall be a sheet metal template with hold-down hardware. Latches shall be provided for engaging with wing-nut fasteners on the receiver-transmitter. The mounting shall be attached to the plane mounting surface by means of 8 machine screws. The overall dimensions of the mounting shall not exceed 9-1/2 inches wide by 1 inch high by 14-1/4 inches long.

3.4.3.3 Weight - The weight of the mounting shall not exceed 0.5 pound.

*3.4.3.4 Safety Wire Hole - The safety wire hole in the mounting shall eliminate the possibility of cutting wire during use.

*3.4.4 Cooler, Air, Electronic Equipment HD-513/ARC - The cooler shall meet the following requirements:

3.4.4.1 Function - The cooler shall provide a circulation means for external air when the radio set is operating in a self-cooling mode.

*3.4.4.2 Form Factor and Mounting - The cooler shall be a single subassembly capable of being attached to the rear of the receiver-transmitter by means of captive screws. It shall be possible to remove the cooler from the receiver-transmitter using ordinary hand tools and without requiring removal of any cables. The overall dimensions of the cooler shall not exceed 6-5/32 inches wide by 6-3/8 inches high by 3-1/8 inches long.

*3.4.4.3 Weight - The weight shall not exceed 1.1 pound.

3.4.4.4 Contents - This cooler shall contain a blower and air transition means for circulating ambient air through the external cover of the heat exchanger on the receiver-transmitter. The cooler shall be powered and controlled by means of a connector mating with the rear connector of the receiver-transmitter.

3.4.4.5 Electrical Connections - All electrical connections to the cooler shall be made through a connector conforming to Specification MIL-C-26482.

3.4.4.6 Vibration - The cooler shall be subjected to vibration only when attached to the receiver-transmitter.

*3.4.5 Indicator, Standing Wave Ratio IM-178/ARC-88 - The standing wave ratio indicator shall meet the following requirements.

*3.4.5.1 Function - The standing wave ratio indicator shall provide a means to measure incident and reflected power of the receiver-transmitter when used with a remote meter and provide a low-pass antenna filter.

*3.4.5.2 Form Factor and Mounting - The standing wave ratio indicator shall be contained in a rectangular box capable of being attached to the front of the receiver-transmitter by means of captive screws. It shall be possible to remove the unit from the receiver-transmitter using ordinary hand tools and without requiring removal of any cables except the pendant cable attached to the antenna jack of the receiver-transmitter. The over-all dimensions shall not exceed 4-3/4 inches wide by 2-3/16 inches high by 3-3/4 inches long.

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*3.4.5.3 Weight - The weight shall not exceed 1.1 pounds.

*3.4.5.4 Contents - The indicator shall contain a connector, a switch, and all circuits necessary to provide incident and reflected power measurements of the receiver-transmitter with a remote meter and to attenuate all emitted antenna signals above the highest carrier frequency.

*3.4.5.5 Electrical Connections - All electrical connections to the indicator shall be made through connectors conforming to Specification MIL-C-26482.

*3.4.5.6 Controls, Panel-Mounted - The following control shall be located and suitably marked on the front panel:

(1) A momentary push-button switch to be depressed for measuring reflected power by means of the remote meter.

*3.4.5.7 Operation - The momentary "Press for Refl Pwr" switch shall not be connected in the internal measuring circuit. Reflected power measurements shall normally be accomplished by operation of a remote switch located near the remote meter. It shall be possible by changing connections within the IM-178/ARC-88 to effect switching of the incident and reflected power measurements by the switch contained therein.

*3.5 Associated Equipment - The equipment shall operate with the following associated equipment, which shall not be supplied as part of this equipment:

Antenna	AT-141A/AR or equivalent with RG-8/U cable
Digital Data Communication Set	AN/ASW-21

3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents referenced in Section 2, unless specified in the contract or order.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

* 4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests

(2) Service Approval Tests

* 4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

* 4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

* 4.2.2 Service Approval Tests - At the completion of the contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered at the contractors expense to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

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4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

*4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

*4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent at the contractors expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

*4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitiative results for all acceptance tests.

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Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10- minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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* 4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

*4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be as required by Specification MIL-R-23094, Procedure I. Equipments selected for reliability assurance tests shall first have passed the individual tests.

(1) Minimum Qualification Lot - The first three (3) successive equipments up to the quantity determined by Figure 2 of Specification MIL-R-23094. Use of the largest possible quantity is recommended to reduce total test time. In the event of qualification lot rejection the minimum quantity to be tested shall be increased to six (6) equipments. When the preceding two (2) test groups each experience a reject decision, regardless of the number of unsatisfactory equipments, the minimum quantity to be tested shall be nine (9) equipments less any number of satisfactory equipments following the last unsatisfactory equipment in the preceding test group.

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(2) Quantity To Be Tested For The Sample Plan - When the preceding production lot has successfully completed testing in accordance with the qualification plan, the quantity to be tested shall be determined by the contractor but shall not be less than 4 nor more than the number of equipments determined in accordance with Figure 2 of Specification MIL-R-23094. Any other quantity will require specific approval of the contracting activity. Use of the largest possible sample quantity is recommended to reduce total test time. The Government Inspector or his duly authorized representative shall select the equipments to be tested at random from the first 10 or 1 1/2 times the quantity to be tested whichever is greater. Failure of a sample plan lot shall require a requalification test in accordance with 4.4.3.(1) before resuming sample plan acceptance, except that equipments shall be selected at random for test from the entire lot by the Government Inspector or his duly authorized representative.

*4.4.3.1 Test Level - The Test Level shall be Test Level III (from Table I of Specification MIL-R-23094).

*4.4.3.2 Accept-Reject Criteria - Figure 12 of Specification MIL-R-23094 and an MTBF of 150 hours shall be used to determine the accept-reject criteria.

*4.4.3.3 Length of Heat Portion of Cycle - The length of the heating cycle, during which time the equipment shall be operated, shall be five hours.

*4.4.3.4 Performance Requirements During Test - The performance characteristics to be measured and the minimum performance requirements to be met during the tests shall be as follows:

Characteristic	Performance requirement
R. F. Power Output	3 db below the limits of 3.3.12
R. F. Output Modulation	3.3.13.2
Receiver Sensitivity	3 db below the limits of 3.3.4
Receiver Data Output	3 db below the limits of 3.3.10

*4.4.3.5 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the acceptance requirement (4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

* 4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

* 4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

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4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

* 4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.) The life test on the radio set control shall consist of operating each control knob for a total of 5000 cycles throughout its complete range.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ±5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 V DC and 115 ±1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Non-repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.6. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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* 6.7 Asterisk - An Asterisk "*" preceding a paragraph number denotes paragraphs where changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the asterisk notations and relationship to the last previous issue.

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